Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 9

Roll No : 495

Name : Dhruv Kakkad

01) Write a function to calculate BMI given mass and height. (BMI = mass/h**2)

In [5]:
```
mass = int(input("Enter Mass:"))
h = int(input("Enter h:"))

def func(mass,h):
    return mass/h**2

func(mass,h)
```

Out[5]: 1.0204081632653061

02) Write a function that add first n numbers.

In [7]:
```
n = int(input("Enter n:"))

def func(n):
    if n==0:
        return 0
    else:
        return n + func(n-1)

func(n)
```

Out[7]: 55

03) Write a function that returns 1 if the given number is Prime or 0 otherwise.

In [25]:
```
n = int(input("Enter n:"))
def isPrime(n):
    for i in range(2,n):
        if n%i==0:
            return 0
    else:
        return 1

isPrime(n)
```

Out[25]: 1

04) Write a function that returns the list of Prime numbers between given two numbers.

In [28]:
```
a = int(input("Enter a:"))
b = int(input("Enter b:"))

def Prime(a,b):
    li = list()
    for i in range(a,b+1):
        if i==1:
            continue
        if isPrime(i)==1:
           li.append(i)
        else:
            pass
    return li

Prime(a,b)
```

Out[28]: [2, 3, 5, 7]

05) Write a function that returns True if the given string is Palindrome or False otherwise.

In [1]:
```
str = input("Enter String:")

def isPalindrone(str):
        if str == str[::-1]:
            return True
        else:
             return False

isPalindrone(str)
```

Out[1]: True

06) Write a function that returns the sum of all the elements of the list.

In [40]:
```
n = int(input("Enter Number:"))

def sumOfElements(n):
    sum = 0
```

```
    for i in range(1,n+1):
        sum +=i

    print(sum)
sumOfElements(n)
```

55

07) Write a function to calculate the sum of the first element of each tuples inside the list.

In [6]:
```
data = [(1, 2), (3, 4), (5, 6)]
def sum_of_first(data):
    sum = 0
    for t in data:
        sum+=t[0]
    print(sum)

sum_of_first(data)
```

9

08) Write a recursive function to find nth term of Fibonacci Series.

In [42]:
```
def fibonacci(n:int)->int:
    if n<=0:
        return 1
    if n==1:
        return 0
    if n==2:
        return 1
    return fibonacci(n-1)+fibonacci(n-2)

print(fibonacci(5))
```

3

09) Write a function to get the name of the student based on the given rollno.

Example: Given dict1 = {101:'Ajay', 102:'Rahul', 103:'Jay', 104:'Pooja'} find name of student whose rollno = 103

In [10]:
```
dict1 = {101:'Ajay', 102:'Rahul', 103:'Jay', 104:'Pooja'}
def checkName(dict1,rollno):
        return dict1[rollno]

print(checkName(dict1,103))
```

Jay

10) Write a function to get the sum of the scores ending with zero.

Example : scores = [200, 456, 300, 100, 234, 678]

Ans = 200 + 300 + 100 = 600

In [11]:
```
li = [200, 456, 300, 100, 234, 678]
def addZero(li):
    sum = 0
    for i in li:
        if i%10==0:
            sum+=i
    print(sum)

addZero(li)
```

```
600
```

11) Write a function to invert a given Dictionary.

hint: keys to values & values to keys

Before : {'a': 10, 'b':20, 'c':30, 'd':40}

After : {10:'a', 20:'b', 30:'c', 40:'d'}

In [51]:
```
d1 = {'a': 10, 'b':20, 'c':30, 'd':40}
inv_di = {}
for k,v in d1.items():
 inv_di[v] = k
print(inv_di)
```

```
{10: 'a', 20: 'b', 30: 'c', 40: 'd'}
```

12) Write a function that returns the number of uppercase and lowercase letters in the given string.

example : Input : s1 = AbcDEfgh ,Ouptput : no_upper = 3, no_lower = 5

In [15]:
```
s1 = "AbcDEfgh"
def upperLower(s1):
    lower=0
    upper=0
    for i in s1:
        if i.islower():
            lower+=1
        else:
            upper+=1
    print(f"no_upper:{upper},no_lower:{lower} ")
upperLower(s1)
```

```
no_upper:3,no_lower:5
```

13) Write a lambda function to get smallest number from the given two numbers.

In [68]:
```
ans = lambda x,y : x < y
ans(1,2)
```

Out[68]:
```
True
```

14) For the given list of names of students, extract the names having more that 7 characters. Use filter().

In [18]:
```
students = ["Ajay", "Rahul", "Pratiksha", "Samantha", "Jay", "Pooja"]

result = list(filter(lambda students: len(students) > 7, students))

print(result)
```

```
['Pratiksha', 'Samantha']
```

15) For the given list of names of students, convert the first letter of all the names into uppercase. use map().

In [21]:
```
students = ["ajay", "aahul", "Aratiksha", "Samantha", "jay", "pooja"]

res = list(map(lambda students:students.title(),students))
print(res)
```

```
['Ajay', 'Aahul', 'Aratiksha', 'Samantha', 'Jay', 'Pooja']
```

16) Write udfs to call the functions with following types of arguments:

1. Positional Arguments
2. Keyword Arguments
3. Default Arguments
4. Variable Legngth Positional(*args) & variable length Keyword Arguments (**kwargs)
5. Keyword-Only & Positional Only Arguments

In [22]:
```
def add(a, b):
    print(a + b)

add(5, 3)

def student(name, age):
    print(name, age)

student(age=20, name="Rahul")


def greet(name="Guest"):
    print("Hello", name)

greet()
greet("Ajay")

def total(*numbers):
    print(sum(numbers))

total(1, 2, 3, 4)

def details(**data):
    print(data)
```

```
details(name="Rahul", age=20)

def pos_only(a, b, /):
    print(a + b)

pos_only(2, 3)

def key_only(*, name, age):
    print(name, age)

key_only(name="Pooja", age=22)
```

```
8
Rahul 20
Hello Guest
Hello Ajay
10
{'name': 'Rahul', 'age': 20}
5
Pooja 22
```

In [ ]: